WORKSHEET — UNIT IV

IoT Security and Data Analytics

Course: IoT with Applications

Unit: IV (3 Hrs / 180 min)

Topics: IoT Security Fundamentals; IoT Threats and Attacks; IoT Data Analytics; Blockchain + IoT; Security Checklist and Analytics Pipeline

🎯 Section A — Learning Objectives

1. Explain the IoT security requirements — confidentiality, integrity and availability — and the roles of authentication, authorization and encryption.

2. Describe secure communication and the security measures needed at the device, network and cloud levels.

3. Identify the major IoT threats — device attacks, eavesdropping, spoofing, DoS, malware and data leakage — and match each with a countermeasure.

4. Describe the IoT data analytics pipeline: preprocessing, time-series data, visualisation, anomaly detection and predictive analytics.

5. Explain how blockchain supports IoT through decentralisation, smart contracts, data integrity and identity management.

6. Apply a security checklist and an analytics pipeline to a real-world IoT scenario.

🔑 Section B — Key Concepts

No.	Concept	Short Explanation
1	**CIA Triad**	Confidentiality (only authorised parties can read data), Integrity (data is not altered) and Availability (services stay reachable) are the three core security goals.
2	**Authentication vs Authorization**	Authentication proves who a user or device is; authorization decides what that verified identity is allowed to do.
3	**Encryption**	Turns data into unreadable ciphertext; symmetric ciphers (e.g. AES) are fast enough for constrained devices, while asymmetric methods (e.g. ECC, RSA) handle key exchange and digital signatures.
4	**Secure Communication**	TLS (over TCP) and DTLS (over UDP) protect data in transit, for example MQTT over TLS, HTTPS and CoAP over DTLS.
5	**Layered Security**	Defence at every level: device (secure boot, signed firmware), network (firewalls, segmentation) and cloud (access control, encryption at rest).
6	**Common IoT Attacks**	Eavesdropping, spoofing, denial of service, malware, device tampering and data leakage each target one or more security goals.
7	**Preprocessing and Time-Series Data**	Raw sensor readings are cleaned (missing values, noise, outliers) and kept as timestamped series before they are analysed.
8	**Anomaly Detection and Predictive Analytics**	Anomaly detection flags unusual readings; predictive analytics forecasts future values or failures, increasingly using machine learning.
9	**Blockchain for IoT**	A shared ledger of hash-linked blocks that is tamper-evident and supports decentralised trust, smart contracts and device identity.

📘 Section C — Important Terminology

Keyword	Meaning
CIA Triad	Confidentiality, Integrity and Availability: the three core goals of information security.

Authentication	Verifying the identity of a user or device, for example with passwords, certificates or tokens.
Authorization	Deciding which resources and actions an authenticated user or device is permitted to use.
Encryption	Transforming data into unreadable ciphertext with a key so that only authorised parties can read it, e.g. AES.
TLS / DTLS	Transport Layer Security (over TCP) and Datagram TLS (over UDP): protocols that encrypt and authenticate data in transit.
Secure Boot	A start-up check that lets a device run only firmware that is digitally signed by a trusted source.
Eavesdropping	Secretly capturing data in transit, such as sniffing unencrypted wireless traffic.
Spoofing	Impersonating a legitimate device, user or address to gain access or to inject false data.
DoS / DDoS	(Distributed) Denial of Service: flooding a device or service with requests so legitimate users cannot reach it.
Malware	Malicious software, such as viruses, ransomware or botnet code, that compromises devices.
Data Leakage	Unintended exposure of sensitive data, for example through weak access control or a misconfigured cloud store.
Time-Series Data	Sensor readings recorded in sequence with timestamps, showing trends, seasonality and noise.
Data Preprocessing	Cleaning and preparing raw data — handling missing values, noise and outliers, and scaling — before analysis.
Blockchain	A distributed ledger of blocks linked by cryptographic hashes, which makes recorded data tamper-evident.
Smart Contract	Self-executing code stored on a blockchain that runs automatically when agreed conditions are met.

✅ Section D — Multiple-Choice Questions [Easy–Medium]

1. Which three goals make up the CIA triad in security?

- **A.** Connectivity, Interoperability, Automation
- **B.** Cloud, Intelligence, Analytics
- **C.** Control, Isolation, Authorization
- **D.** Confidentiality, Integrity, Availability

2. Which process verifies that a device really is who it claims to be?

- **A.** Authorization
- **B.** Authentication
- **C.** Encryption
- **D.** Compression

3. Which protocol is used to protect CoAP traffic that runs over UDP?

- **A.** DTLS
- **B.** SSH
- **C.** FTP
- **D.** SNMP

4. An attacker captures unencrypted Wi-Fi packets from a smart sensor to read its data. What is this attack called?

A. Spoofing
B. Denial of Service
C. Eavesdropping
D. Ransomware

5. Which attack floods an IoT service with traffic so that legitimate users cannot reach it?

A. Spoofing
B. Denial of Service (DoS)
C. Data leakage
D. Secure boot

6. Which device-security feature allows only digitally signed firmware to run at start-up?

A. Firewall
B. Time-series database
C. Smart contract
D. Secure boot

7. Which analytics step handles missing values, noise and outliers before the data is analysed?

A. Data preprocessing
B. Visualisation
C. Encryption
D. Authorization

8. What describes time-series data?

A. Only text labels
B. Only images
C. Readings recorded in sequence with timestamps
D. Only encrypted values

9. Which technology provides a tamper-evident ledger by linking blocks with cryptographic hashes?

A. Fog computing
B. Blockchain
C. Bluetooth
D. Time-series database

10. What is a smart contract?

A. Self-executing code on a blockchain that runs when agreed conditions are met
B. A paper agreement signed by the device owner
C. A type of firewall rule
D. A cloud billing plan

✏️ Section E — Fill in the Blanks [Easy]

1. _____ decides what an authenticated user or device is allowed to do.

2. _____ converts readable data into unreadable ciphertext using a key.

3. A DDoS attack often uses a large network of infected devices, called a _____.

4. _____ analytics uses historical sensor data to forecast future values or failures.

5. In a blockchain, each block stores the _____ of the previous block, which makes tampering easy to detect.

☑️ Section F — True or False [Easy]

1. Changing the default password of an IoT device improves its security. (True / False)

2. Encryption alone guarantees that a device is secure, so no other measures are needed. (True / False)

3. Spoofing means impersonating a legitimate device or user. (True / False)

4. A blockchain stores all data on one central server controlled by a single organisation. (True / False)

5. Raw sensor data can be fed straight into analytics without any preprocessing. (True / False)

🔗 Section G — Match the Following [Medium]

Column A	Column B (shuffled)
1. Authentication	a. Self-executing code on a blockchain that runs when conditions are met
2. Authorization	b. Impersonating a legitimate device or user
3. Spoofing	c. Allowing a device to run only trusted, digitally signed firmware
4. Data leakage	d. Verifying who a user or device is
5. Secure boot	e. Flagging sensor readings that deviate from normal behaviour
6. Smart contract	f. Deciding what a verified user or device may access
7. Anomaly detection	g. Unintended exposure of sensitive data

📝 Section H — Short-Answer Questions [Medium]

1. Differentiate between authentication and authorization, with an IoT example.

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2. What is the CIA triad? Name its three goals.

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3. How do TLS and DTLS protect IoT communication?

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4. Why is data preprocessing necessary before analysing IoT sensor data? Give two preprocessing steps.

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5. How does a blockchain make recorded IoT data tamper-evident?

📄 Section I — Descriptive Questions [Medium–Advanced]

1. Explain the IoT security requirements and describe security at the device, network and cloud levels, including authentication, authorization, encryption and secure communication.

2. Describe the main IoT threats — device attacks, eavesdropping, spoofing, DoS attacks, malware and data leakage — and give one countermeasure for each.

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3. Explain the IoT data analytics pipeline — preprocessing, time-series data, visualisation, anomaly detection, predictive analytics and AI/ML-based analytics — with a diagram.

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4. Explain how blockchain can support IoT through decentralisation, smart contracts, data integrity and identity management, and discuss its limitations for constrained devices.

💻 Section J — Practical / Application-Based Questions [Advanced]

Practical 1

Problem: Use Python's hmac and hashlib modules to protect a sensor message against tampering.

Task:

1. Create a shared secret key and the message "temp=28".

2. Compute the HMAC-SHA256 tag of the message.

3. Verify the original message, then verify the tampered message "temp=18" against the same tag, and print both results.

Expected Output: *Two lines: "Original valid: True" and "Tampered valid: False".*

Practical 2

Problem: Identify the attack in each scenario and suggest a suitable countermeasure.

Task:

1. A neighbour captures unencrypted Wi-Fi packets from a baby-monitor camera.

2. Thousands of infected cameras flood a company's server with requests until it stops responding.

3. An attacker connects a fake sensor that sends false temperature readings to the cloud.

Expected Output: *A table giving the attack, the security goal affected and one countermeasure for each scenario.*

Practical 3

Problem: Clean a small sensor series and smooth it with a moving average, as in the preprocessing step of an analytics pipeline.

Task:

1. Store the readings [20, 21, None, 23, 22, 24] in a list, where None marks a missing value.

2. Replace each missing value with the previous valid reading.

3. Compute the 3-point moving average and print the results rounded to two decimal places.

Expected Output: *The cleaned list [20, 21, 21, 23, 22, 24] and the moving averages [20.67, 21.67, 22.0, 23.0].*

🗺️ Section K — Concept-Mapping Activity

IOT SECURITY AND DATA ANALYTICS

Complete the map below: fill in the missing sub-topic name or example for each branch.

Branch 1	Branch 2	Branch 3	Branch 4
Security Fundamentals	**Threats and Attacks**	**Data Analytics**	**Blockchain + IoT**
Example: _____	Example: _____	Example: _____	Example: _____

(Answer key — example values: authentication, authorization, encryption, TLS; eavesdropping, spoofing, DoS; preprocessing, time-series, anomaly detection; smart contracts, data integrity, identity management)

🏢 Section L — Case Study / Real-World Activity

Scenario

A university installs 150 IP cameras and smart door locks across its campus. The devices still use the factory-default password, their firmware has never been updated, and video is streamed unencrypted to a cloud dashboard. Recently several cameras became slow, and the campus network went down when a flood of requests hit an internal server. Traffic analysis showed that some cameras had been infected and were sending the requests.

Questions

1. Identify three security weaknesses in this deployment.

2. Which attacks have occurred or are now possible? Name them.

3. Recommend countermeasures at the device, network and cloud levels.

4. How could anomaly detection on network traffic, and blockchain-based device identity, help this university?

Expected Learning Outcome

Students apply a security checklist, knowledge of common threats and basic analytics ideas to harden a realistic IoT deployment.

📊 Section M — Self-Assessment

☐ **Yes** ☐ **Partly** ☐ **No** I can explain the major concepts of this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can define the important terminology from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can solve problems/exercises from this unit.

☐ **Yes** ☐ **Partly** ☐ **No** I can apply the concepts of this unit to a new situation.

💭 Section N — Student Reflection

What did I learn in this unit?

What was difficult for me?

What do I need to practice more?

One important concept I learned:

🗝 Section O — Instructor Answer Key

(For instructor use — covers all objective and subjective items in this worksheet.)

MCQ Answers

- 1. D — Confidentiality, Integrity and Availability are the three core goals of information security.
- 2. B — Authentication verifies identity; authorization only decides what a verified identity may do.
- 3. A — DTLS adds TLS-style encryption and authentication to datagram (UDP) traffic such as CoAP.
- 4. C — Capturing unencrypted traffic to read it is eavesdropping (sniffing), an attack on confidentiality.
- 5. B — A DoS attack overwhelms a service with traffic, attacking availability.
- 6. D — Secure boot checks the firmware's digital signature before letting it run.
- 7. A — Preprocessing cleans missing values, noise and outliers so later analysis is reliable.
- 8. C — Time-series data is a sequence of timestamped readings, showing trends and seasonality.
- 9. B — Blockchain links blocks with cryptographic hashes, so altering one block breaks the chain.
- 10. A — A smart contract is self-executing code on a blockchain that runs when its conditions are met.

Fill in the Blanks — Answers

- 1. Authorization
- 2. Encryption
- 3. botnet
- 4. Predictive
- 5. hash

True/False — Answers

- 1. True
- 2. False
- 3. True
- 4. False
- 5. False

Match the Following — Answer Key

1 → d 2 → f 3 → b 4 → g 5 → c 6 → a 7 → e

Short Answer — Key Points

- 1. Authentication proves identity (e.g. a smart lock checks a user's credential or a sensor presents a certificate); authorization then limits what that identity may do (e.g. the user may unlock only their own door; the sensor may only publish to its own topic).

- 2. Confidentiality (only authorised parties can read data), Integrity (data cannot be altered undetected) and Availability (systems and data stay accessible when needed).
- 3. TLS (over TCP) and DTLS (over UDP) authenticate the endpoints and encrypt the data in transit, protecting against eavesdropping and tampering — e.g. MQTT over TLS and CoAP over DTLS.
- 4. Raw sensor data contains missing values, noise, duplicates and outliers that distort results. Any two steps: handling missing values, filtering noise or smoothing, removing outliers, scaling/normalising, resampling.
- 5. Each block holds the hash of the previous block, so changing any recorded value changes its hash and breaks every later link; the ledger is also copied across many nodes, so a single altered copy is rejected.

Descriptive Questions — Key Points

- 1. Requirements: confidentiality, integrity, availability, plus authentication and authorization. Device: secure boot, signed firmware and OTA updates, unique credentials. Network: firewalls, segmentation, intrusion detection. Cloud: access control, encryption at rest, secure APIs, logging. Communication: TLS/DTLS. Encryption: symmetric (AES) for data, asymmetric (ECC/RSA) for key exchange and signatures. Present as a layered table.
- 2. Device attacks → secure boot, tamper resistance, strong credentials. Eavesdropping → encryption (TLS/DTLS). Spoofing → certificate-based authentication. DoS/DDoS → rate limiting, firewalls, patched devices. Malware → signed updates, network segmentation, monitoring. Data leakage → least-privilege access control and encryption at rest.
- 3. Pipeline: sensors → collection → preprocessing (missing values, noise, outliers, scaling) → time-series storage → visualisation (dashboards, trend charts) → anomaly detection → predictive analytics → AI/ML models → decisions and alerts. Descriptive analytics says what happened; predictive says what will happen.
- 4. Decentralised IoT removes a single point of failure; smart contracts automate agreements (e.g. release payment when a delivery sensor confirms); hash-linked blocks protect data integrity; device identities (public keys) can be registered on the ledger. Limitations: low throughput, storage and energy cost, latency, and constrained devices cannot run full nodes — so use gateways or light clients and store only hashes on-chain.

Practical Activities — Expected Outputs

- 1. Sample logic: import hmac, hashlib; key = b“shared-secret”; msg = b“temp=28”; tag = hmac.new(key, msg, hashlib.sha256).digest(); def ok(m, t): return hmac.compare_digest(hmac.new(key, m, hashlib.sha256).digest(), t). ok(msg, tag) prints True; ok(b“temp=18”, tag) prints False. Note: HMAC gives integrity and authenticity, not confidentiality.
- 2. (1) Eavesdropping — confidentiality — encrypt traffic (TLS, WPA2/WPA3). (2) DDoS from a botnet — availability — rate limiting/DDoS protection, patch devices and change default passwords. (3) Spoofing — integrity/authentication — certificate-based device authentication.
- 3. Sample logic: r = [20, 21, None, 23, 22, 24]; fill each None with the last valid value to get [20, 21, 21, 23, 22, 24]; then for each window of 3, average it: (20+21+21)/3 = 20.67, (21+21+23)/3 = 21.67, (21+23+22)/3 = 22.0, (23+22+24)/3 = 23.0.

Concept Map — Sample Answers

- Security Fundamentals: authentication, authorization, encryption, TLS/DTLS, secure boot. Threats and Attacks: eavesdropping, spoofing, DoS, malware, data leakage. Data Analytics: preprocessing, time-series, visualisation, anomaly detection, predictive analytics. Blockchain + IoT: decentralised IoT, smart contracts, data integrity, identity management.

Case Study — Key Points

- 1. Any three: factory-default passwords (weak authentication); firmware never updated (known vulnerabilities, no secure boot/OTA); unencrypted video (confidentiality); no network segmentation or monitoring.
- 2. Malware (botnet infection) and DoS/DDoS have occurred; eavesdropping on the video stream, spoofing and unauthorised control of door locks are also possible.
- 3. Device: unique strong credentials, secure boot, signed OTA updates. Network: put cameras on a separate VLAN, firewall, rate limiting, intrusion detection. Communication: TLS-encrypted video. Cloud: least-privilege access, multi-factor authentication, encryption at rest, logging.
- 4. Anomaly detection learns normal traffic per camera and flags sudden spikes or unusual destinations so infected devices are isolated early. A shared ledger of device public-key identities lets the system reject spoofed devices and keeps logs tamper-evident, though scalability and cost must be considered.